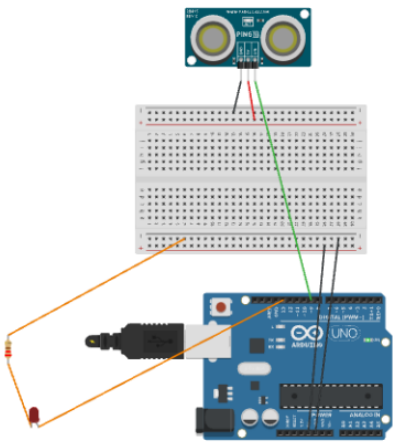
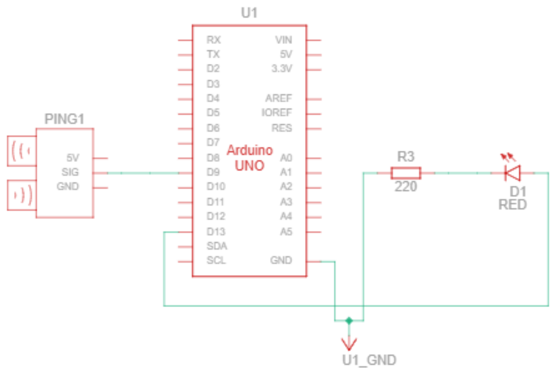
Hellas Planitia Anomaly Survival and Prototyping:

Name: VUCHITHRA VARSHINI K
Roll Number: ME25B1021
Department: Mechanical Engineering
Institution: IIITDM Kancheepuram

Module 1: Navigation Blindspot (Obstacle Avoidance)





Circuit Link:

https://www.tinkercad.com/things/5laKzGcGr6d-module-1navigation-blindspot?sharecode=CFYKq24ctsL7laGlzCEzQc-c9Tg6FuhxxENL5XiroLA

Components

- Arduino Uno R3
- Ultrasonic Sensor
- Red LED
- 220Ω Resistor
- Jumper Wires

Connections

- Ultrasonic VCC → 5V (Arduino)
- Ultrasonic GND → GND (Arduino)
- TRIG → Digital Pin 9
- ECHO → Digital Pin 9 (single signal pin mode)
- LED anode (+) → Digital Pin 13 (through 220Ω resistor)

- LED cathode (–) → GND

How the System Works

The ultrasonic sensor sends a sound pulse from the rover. When this pulse hits an object, it returns back as an echo. Arduino measures the time taken for this return signal and converts it into distance.

If the object is closer than 100 cm, the LED turns ON to indicate an obstacle warning. If the object is farther, the LED remains OFF.

Logic Used

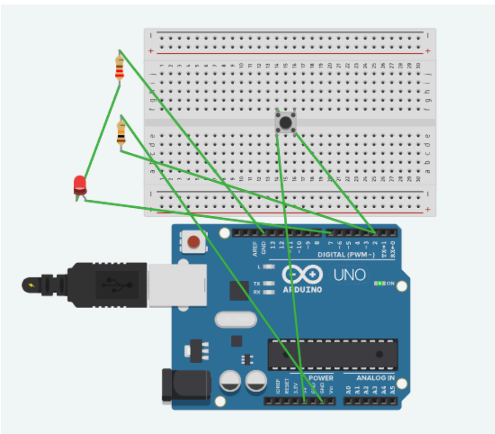
if(distance < 100)

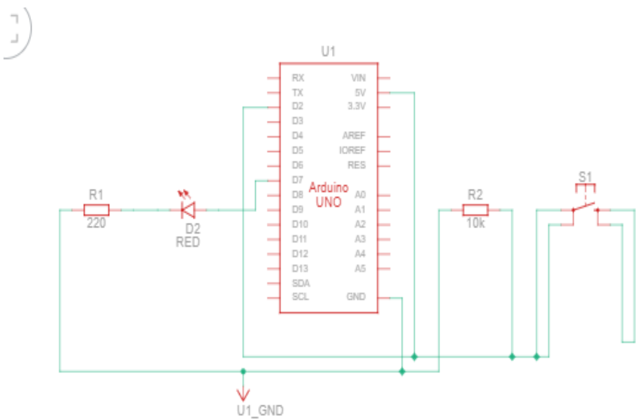
This condition checks whether an obstacle is within danger range.

Engineering Brief

Ultrasonic sensors depend on sound waves, which do not travel efficiently in Mars' thin atmosphere. Dust storms further reduce accuracy. Hence, real rover systems use LiDAR, radar, stereo cameras, IMU, and wheel encoders for reliable navigation.

Module 2: Mobility Trap (Stall Detection)





Circuit Link:

https://www.tinkercad.com/things/9Enl3oDNQRA-module-2mobility-trap

Components Used

- Arduino Uno R3
- Pushbutton Switch
- LED
- 220Ω Resistor
- Jumper Wires

Connections

- Arduino Pin 7 → LED (+) long leg
- LED (-) short leg → 220Ω resistor → GND
- Arduino 5V → Push button (one side)
- Other side of Push button → Arduino Pin 2
- Pin 2 side → 10kΩ resistor → GND (pull-down)

How the System Works

The pushbutton simulates a wheel stall in the rover. When the rover wheel gets stuck in Martian soil, a mechanical limit switch would trigger a HIGH signal.

When the button is pressed, Arduino detects HIGH input and turns ON LED, indicating motor stall condition. When released, the LED turns OFF, showing normal operation.

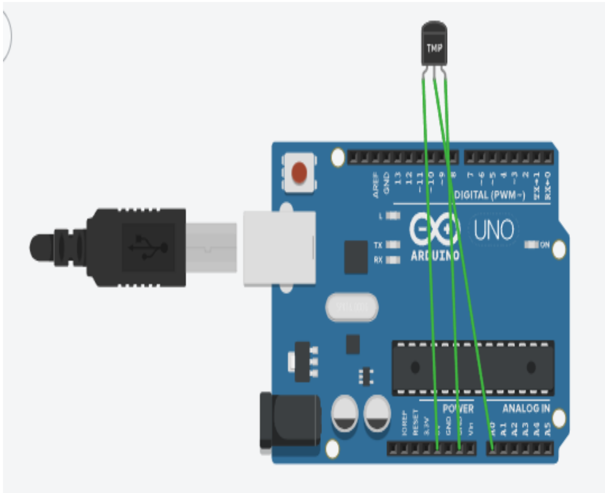
Logic Used

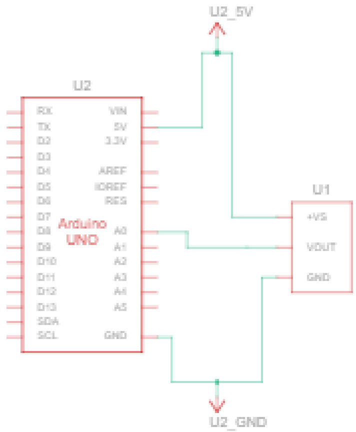
if(buttonState == HIGH)
This detects whether the stall condition is active.

Engineering Brief

Real rover systems detect motor stall using current sensors and wheel encoders. If motor current increases but wheel movement decreases, a stall is detected. The system immediately shuts down motor power to prevent overheating and hardware damage.

Module 3:The Deep Freeze (Thermal Monitoring)





Circuit Link:

https://www.tinkercad.com/things/l1LlkC4MMM3-module-3deep-freeze?sharecode=XN3uyFYVpoJqcht3flwlxDZzb6kQ10KNe5v2yXjFVgQ

Components Used

- Arduino Uno R3
- TMP36 Temperature Sensor
- Jumper Wires

Connections

- TMP36 VCC → 5V
- TMP36 GND → GND
- TMP36 OUT → Analog Pin A0

How the System Works

The TMP36 sensor continuously measures temperature and sends analog voltage signals to Arduino. Arduino converts this signal into temperature values.

The temperature is displayed in real-time on the Serial Monitor every 1 second, allowing continuous thermal monitoring of the rover environment.

Serial Functions Used

- Serial.begin(9600) → starts communication
- Serial.print() → displays sensor data

Engineering Brief

Mars dust storms block solar energy, forcing the rover into low-power survival mode. Only essential systems like thermal control remain active. When temperature drops below safe limits, survival heaters are automatically activated to protect batteries and electronics.

Tinkercad Links

- Module1:https://www.tinkercad.com/things/5laKzGcGr6d-module-1navigation-blindspot?sharecode=CFYKq24ctsL7laGlzCEzQc-c9Tg6FuhxxENL5XiroLA
-
- Module2:https://www.tinkercad.com/things/9Enl3oDNQRA-module-2mobility-trap
- Module3:.https://www.tinkercad.com/things/l1LlkC4MMM3-module-3deep-freeze?sharecode=XN3uyFYVpoJqcht3flwlxDZzb6kQ10KNe5v2yXjFVgQ

System Overview

- Obstacle detection is achieved using an ultrasonic sensor
- Stall detection is implemented using switch-based logic
- Thermal monitoring is performed using a TMP36 temperature sensor
- All subsystems are integrated for real-time decision-making
- Ensures protection and reliability of rover operations
- Supports safe functioning in extreme Martian conditions
- Handles challenges like dust storms, mechanical failure, and severe cold